

Residential Segregation, School Contexts, and Unequal Life Chances

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ROCKWOOL FONDEN

Overblik

1. Kort om min forskning
2. Muslimske skoler i Danmark
3. Diskussion

Uddannelse og ulighed

- **1. Skolevalg og “kvalitet”**
 - *Formelle modeller: kvantitative vs kvalitative skoleeffekter*
 - *Ændringer i skoledistrikter*



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- **1. Skolevalg og “kvalitet”**
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- **2. Hvordan påvirker lærere elevers færdigheder?**
 - *Lærerkvalifikationer vs Lærerbidrag*
 - *Effekter på læring versus ikke-akademiske færdigheder*



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- **3. Etniske sammensætningseffekter**
 - *Påvirker flygtningebørn de øvrige børns læring?*
 - *Indkomstmobilitet og uddannelsesvalg*
 - *Muslimske skoler*



Muslimske skoler

- 20% af danske børn går på friskoler (14% af ikke-vestlige)
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- Ophedet offentlig og politisk debat



Rosenkrantz-Theil slår fast: Vil stadig lukke muslimske friskoler

FRISKOLER: Det er blot coronakrisens skyld, at regeringen ikke er nået videre med ønsket om at tage statsstøtten fra muslimske friskoler. Det fastslog børne- og undervisningsministeren onsdag.

Analyse|kraka

Børn fra muslimske friskoler – hvordan klarer de sig?

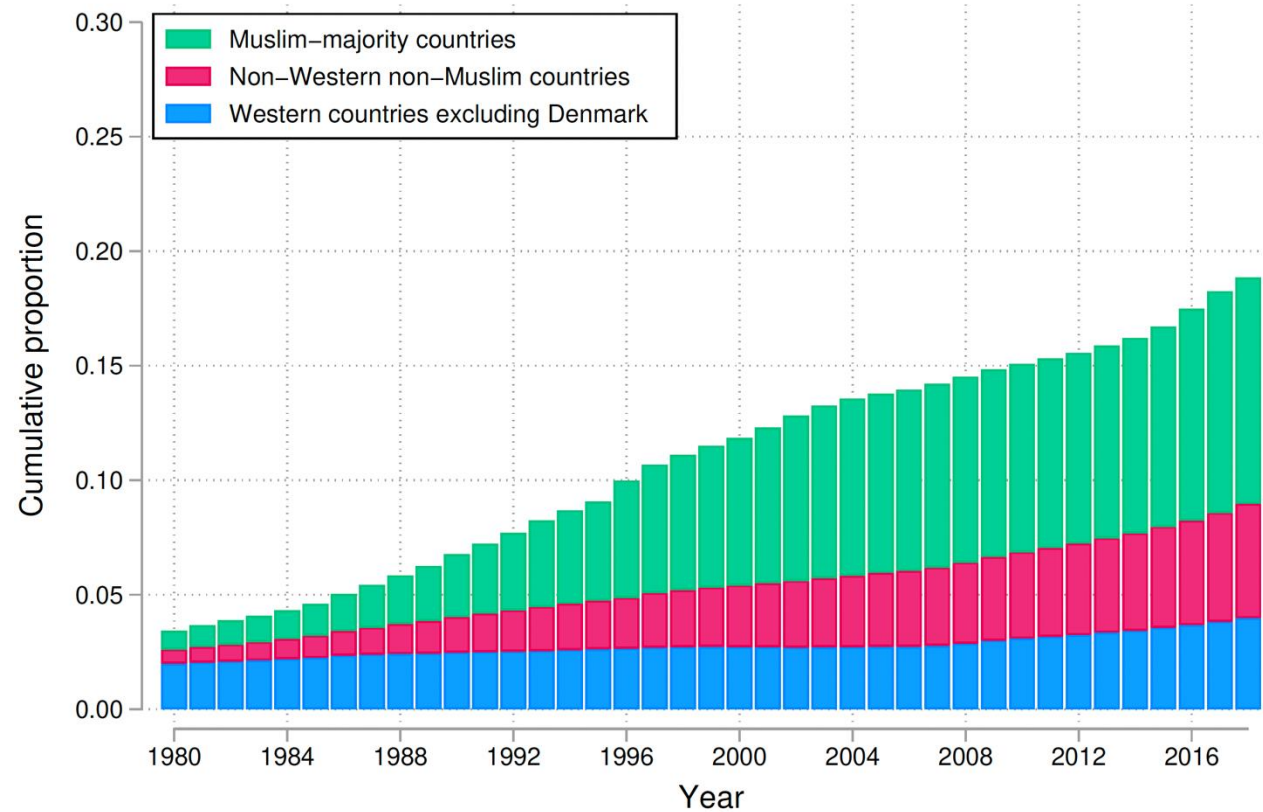
POLITIK

Mette Frederiksen i brutalt opgør: »Luk alle muslimske friskoler«

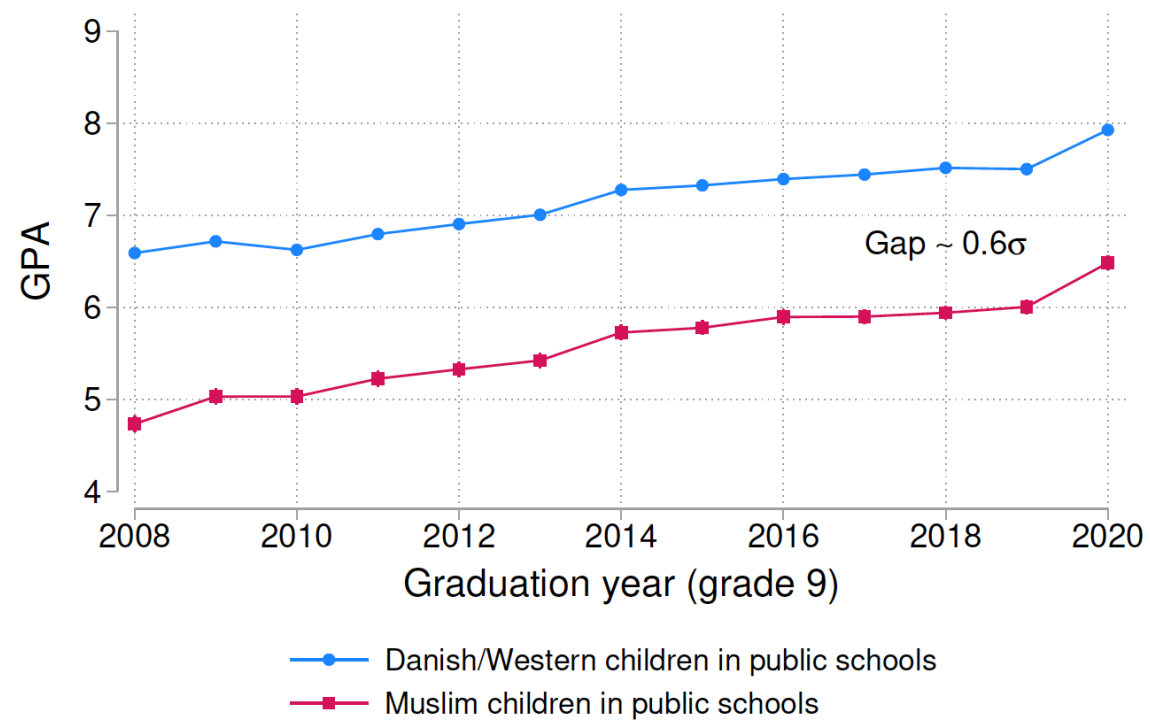
Socialdemokratiets formand Mette Frederiksen vil tage et opgør med de muslimske friskoler i Danmark.



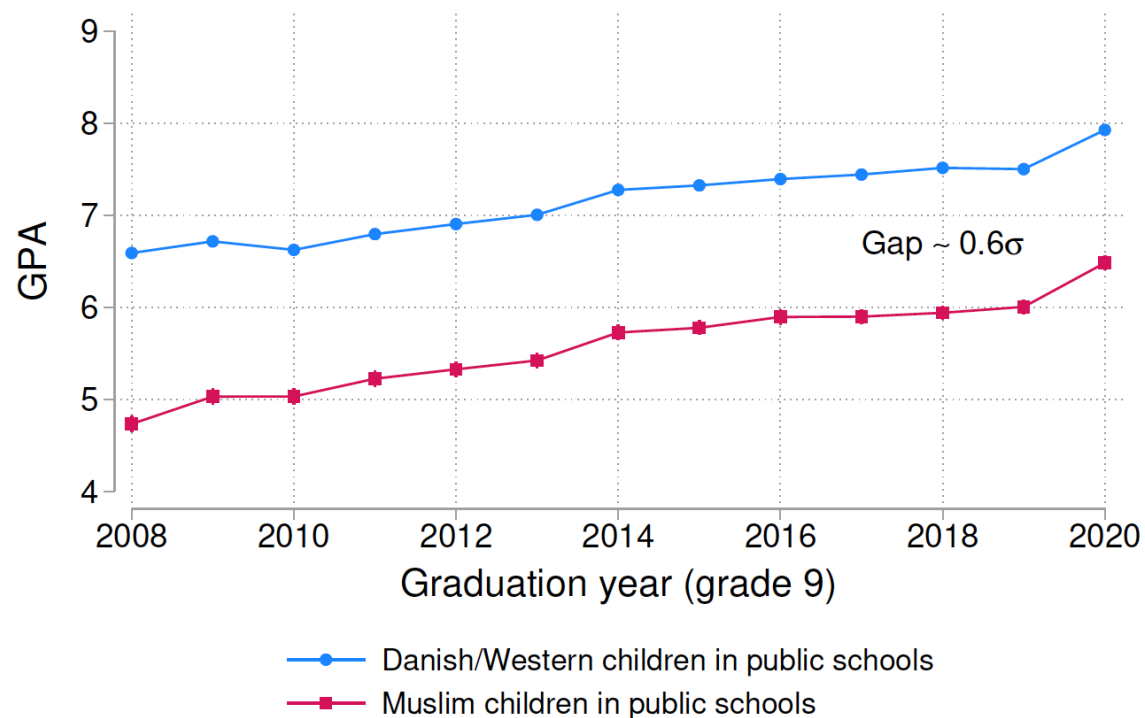
Andel af indvandrere og efterkommere blandt skolebørn (5–15 år) i Danmark, efter oprindelse og år.



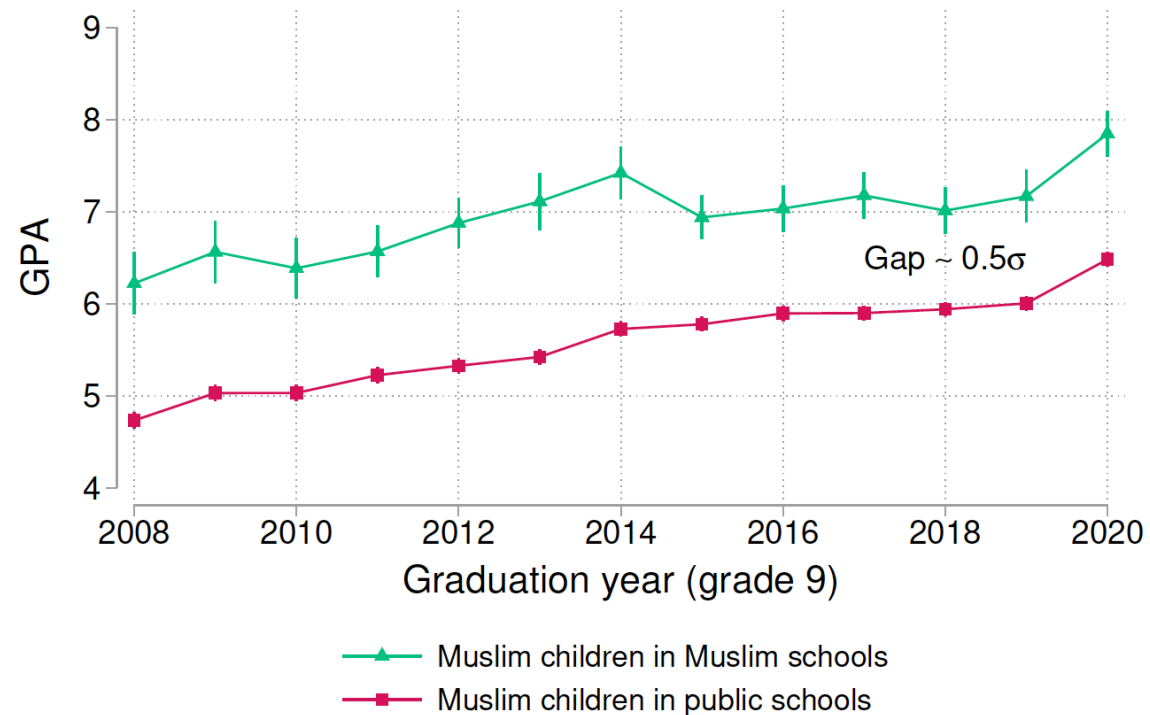
(a) Folkeskoler: Muslimske vs ikke-muslimske børn



(a) Folkeskoler: Muslimske vs ikke-muslimske børn



(b) Folkeskoler vs muslimske friskoler: Muslimske børn



Forskningsspørgsmål

1. Hvordan påvirker muslimske skoler minoritetsbørns læring?

- Naturligt eksperiment: kontrafaktisk sammenligning

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- Naturligt eksperiment: kontrafaktisk sammenligning

2. Hvad forklarer effekterne?

- Lærerbias?
- Etniske lighedseffekter?

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3. Trade off mellem læring og integration?

Studiet

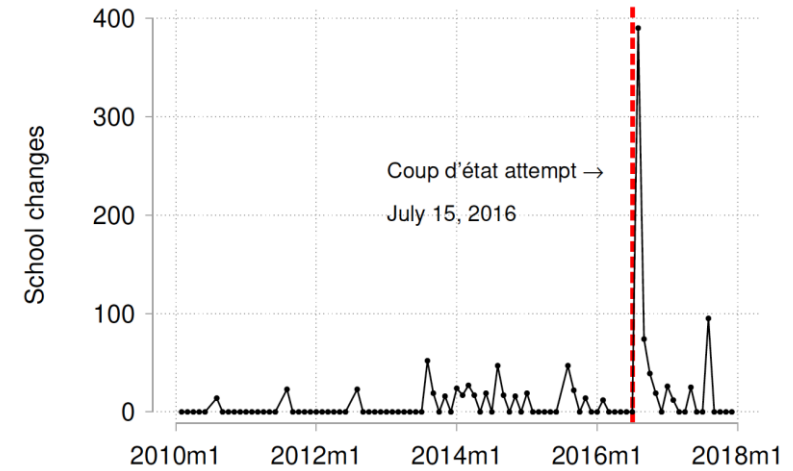
Naturligt eksperiment:

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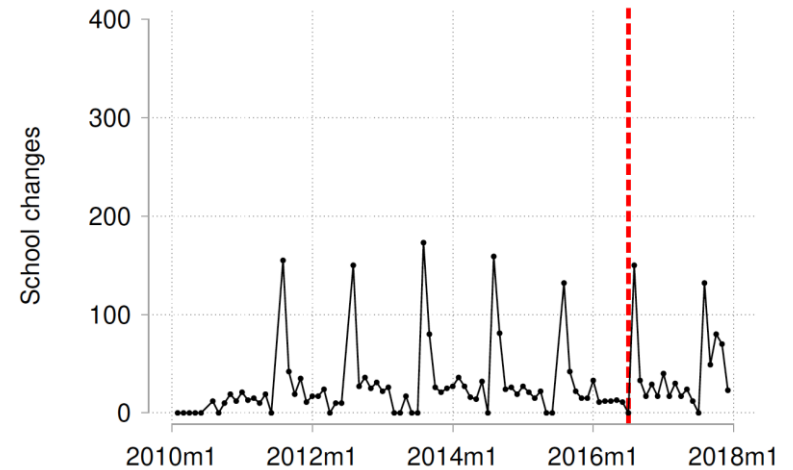
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(a) Muslim Gülen → Public school



(b) Muslim (non-Gülen) → Public

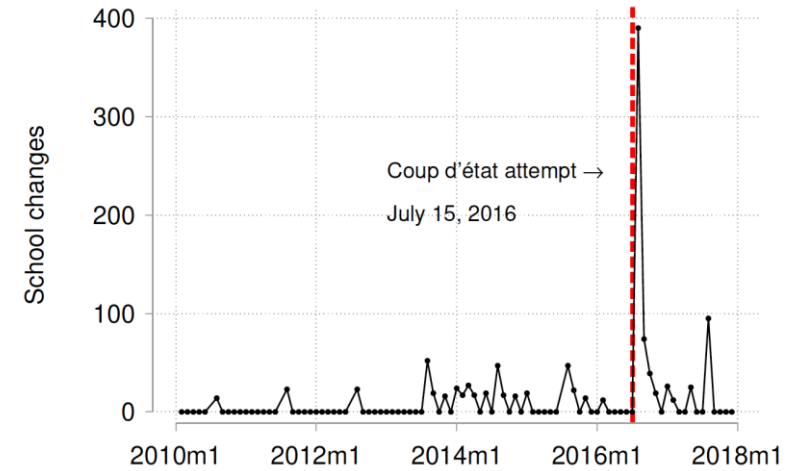
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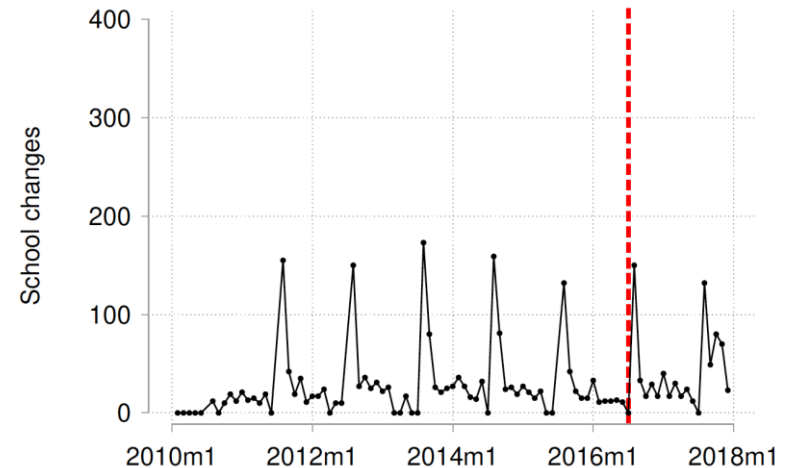
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Sammenligningsgrupper:

- "Heldige elever" afsluttede *før* kupforsøget
- "Uheldige elever" skiftede til folkeskolen
- Ingen forskelle i baggrundsvariable mellem de to grupper



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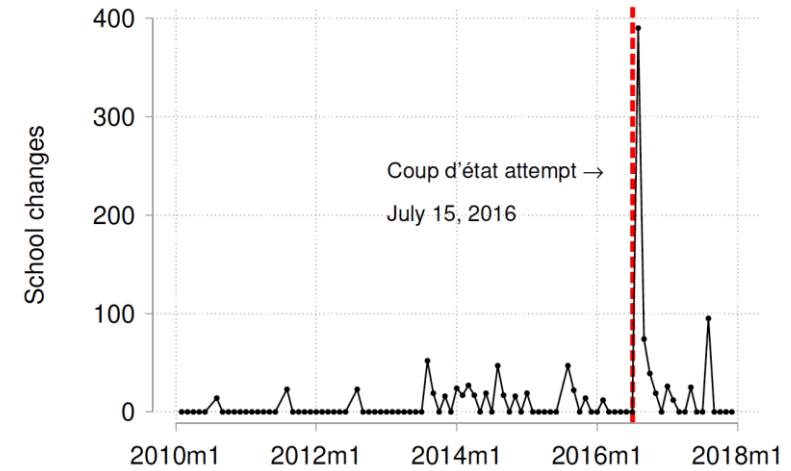
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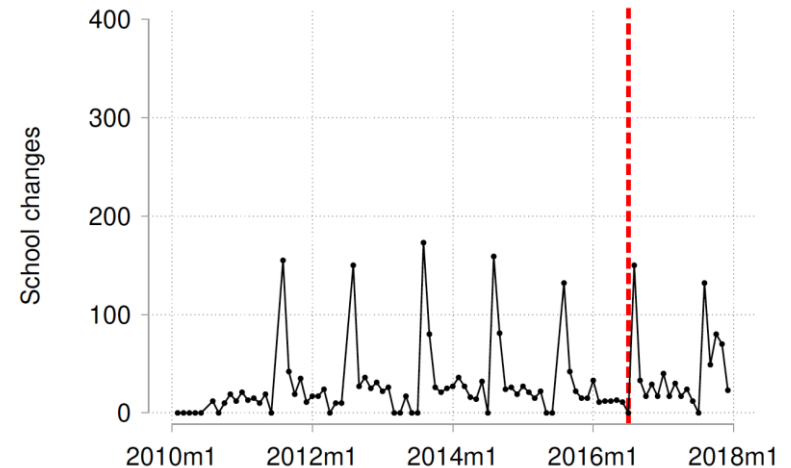
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Forskel i test scores skyldes:

- (A) Forskelle mellem skoletyper
- (B) Forstyrrende effekter ved skoleskift

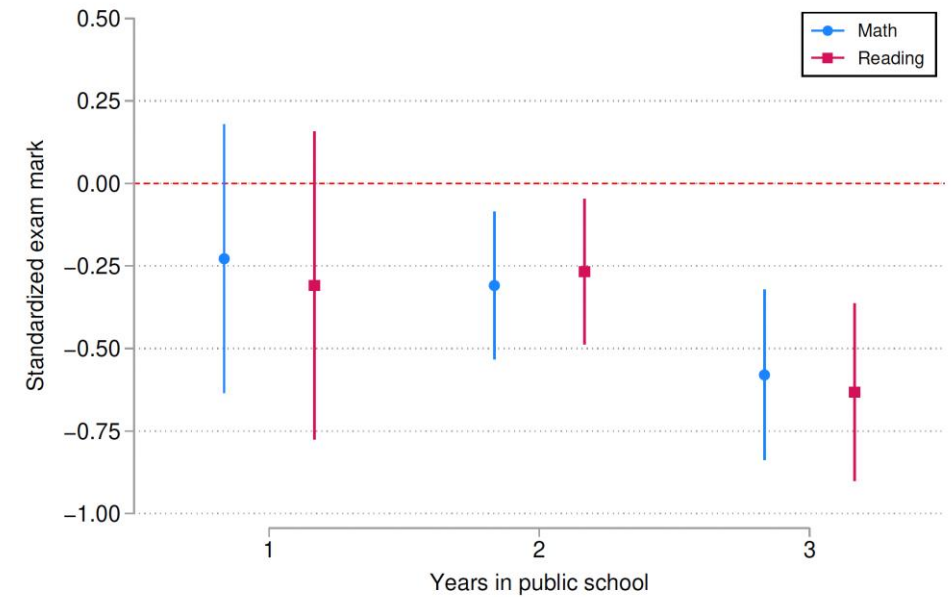
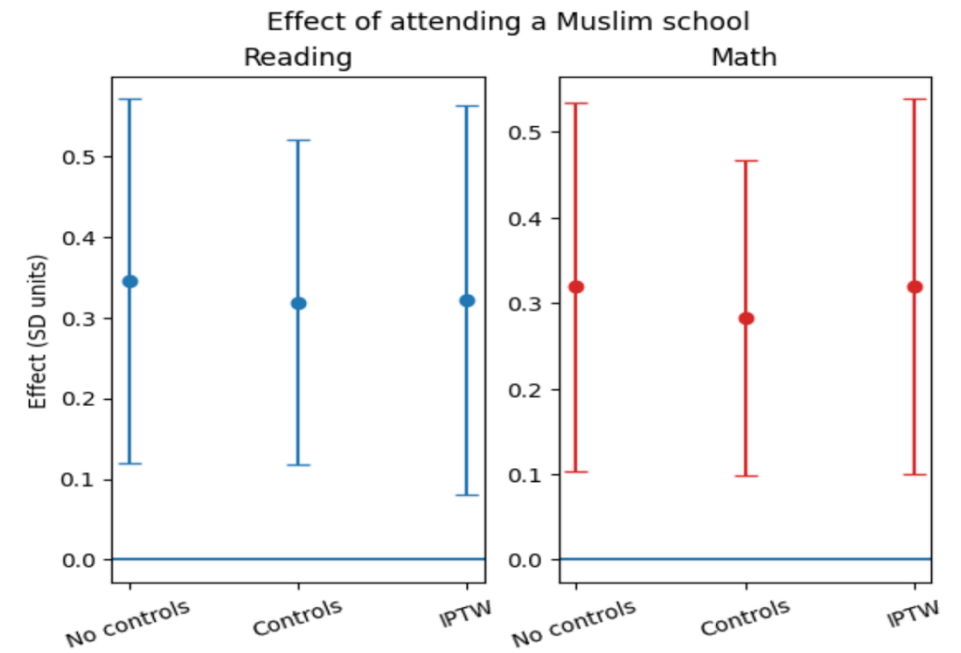


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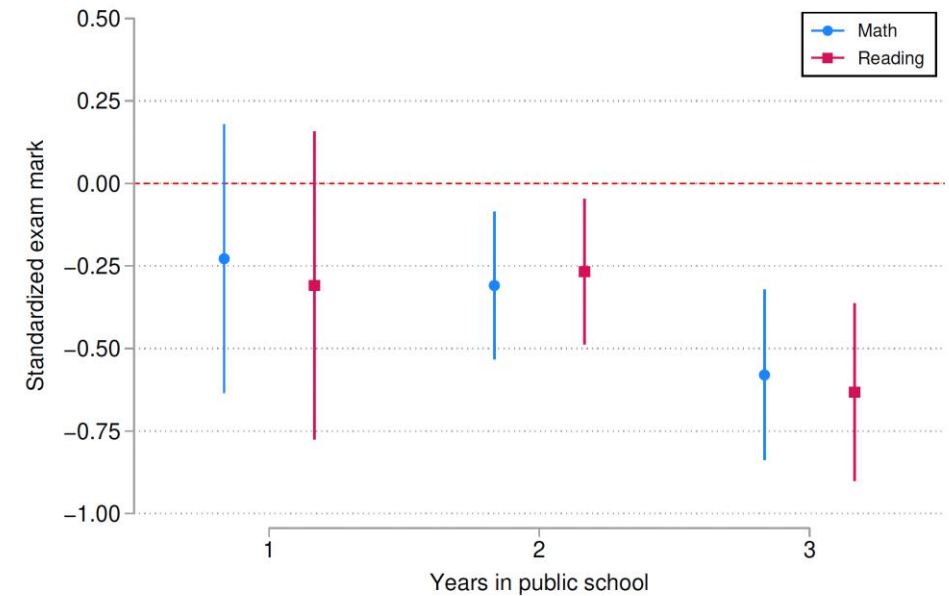
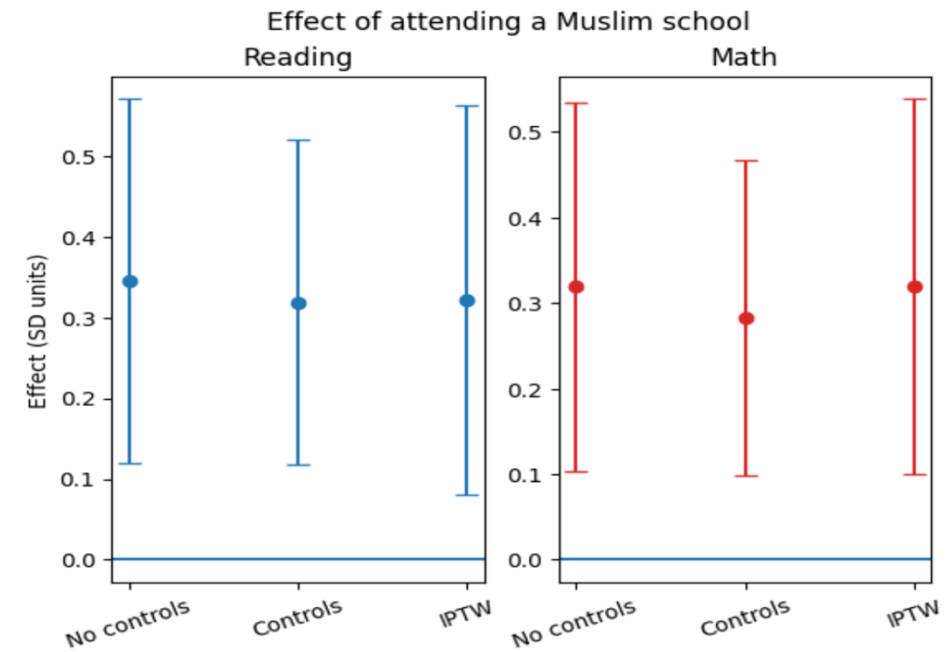
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Hovedresultater



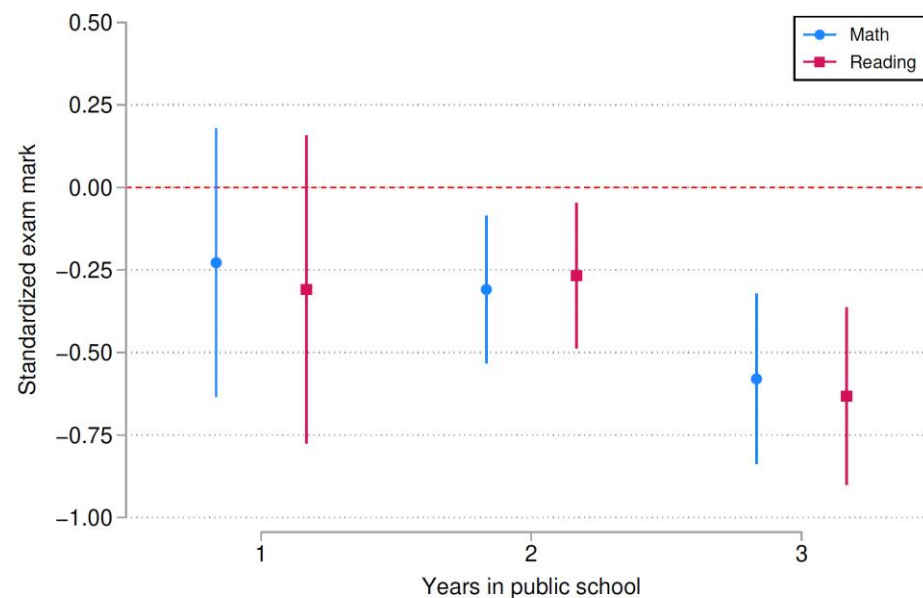
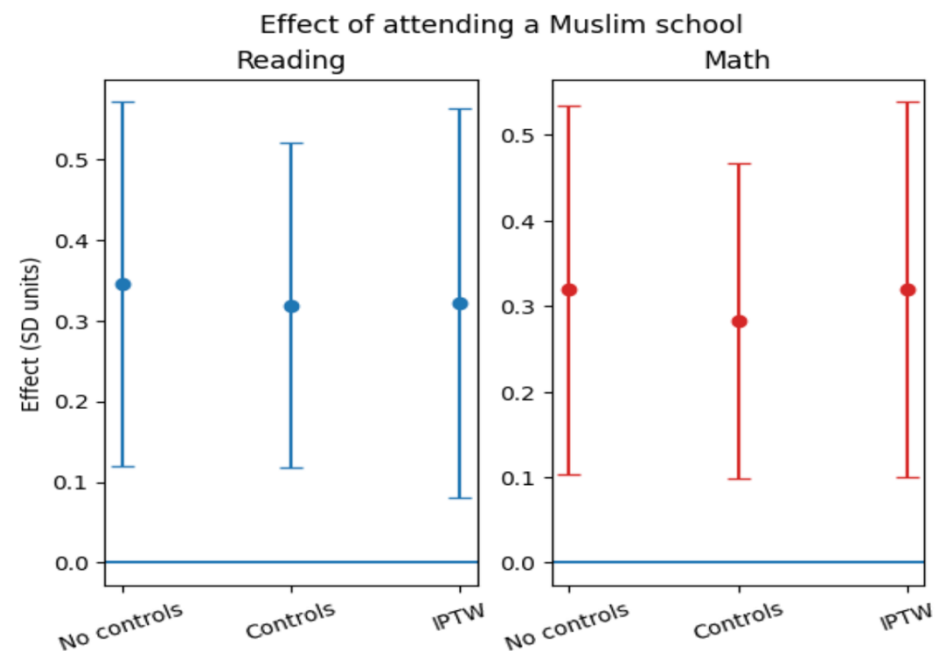
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- **30% SD højere testresultater i dansk og matematik**
- ~ 1 års læring i et ”normalt skoleår”



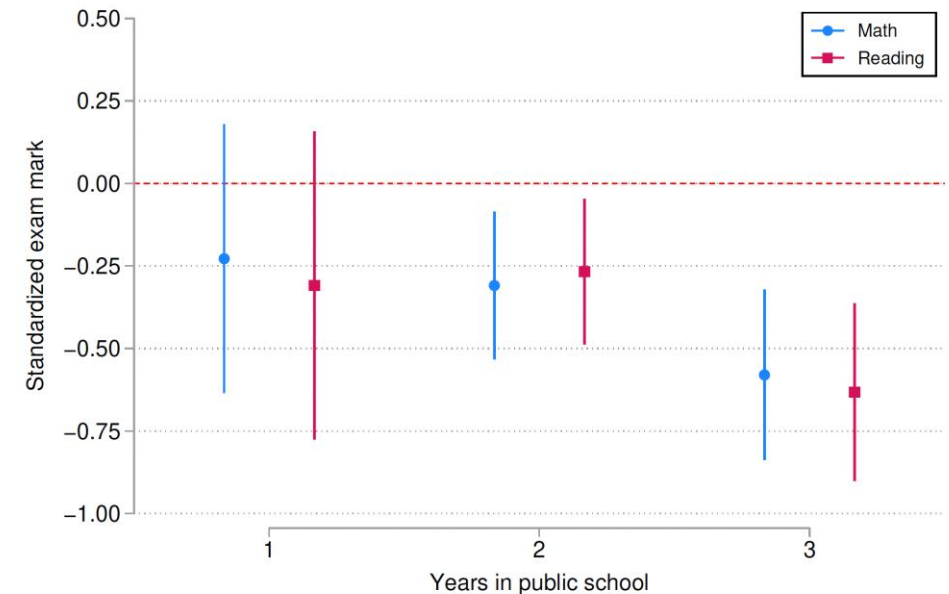
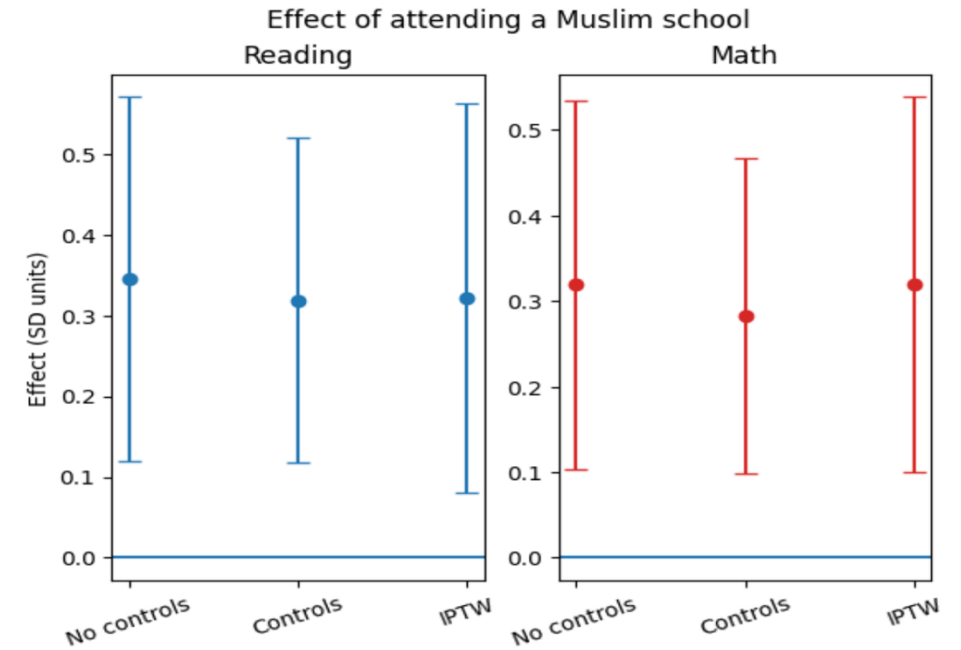
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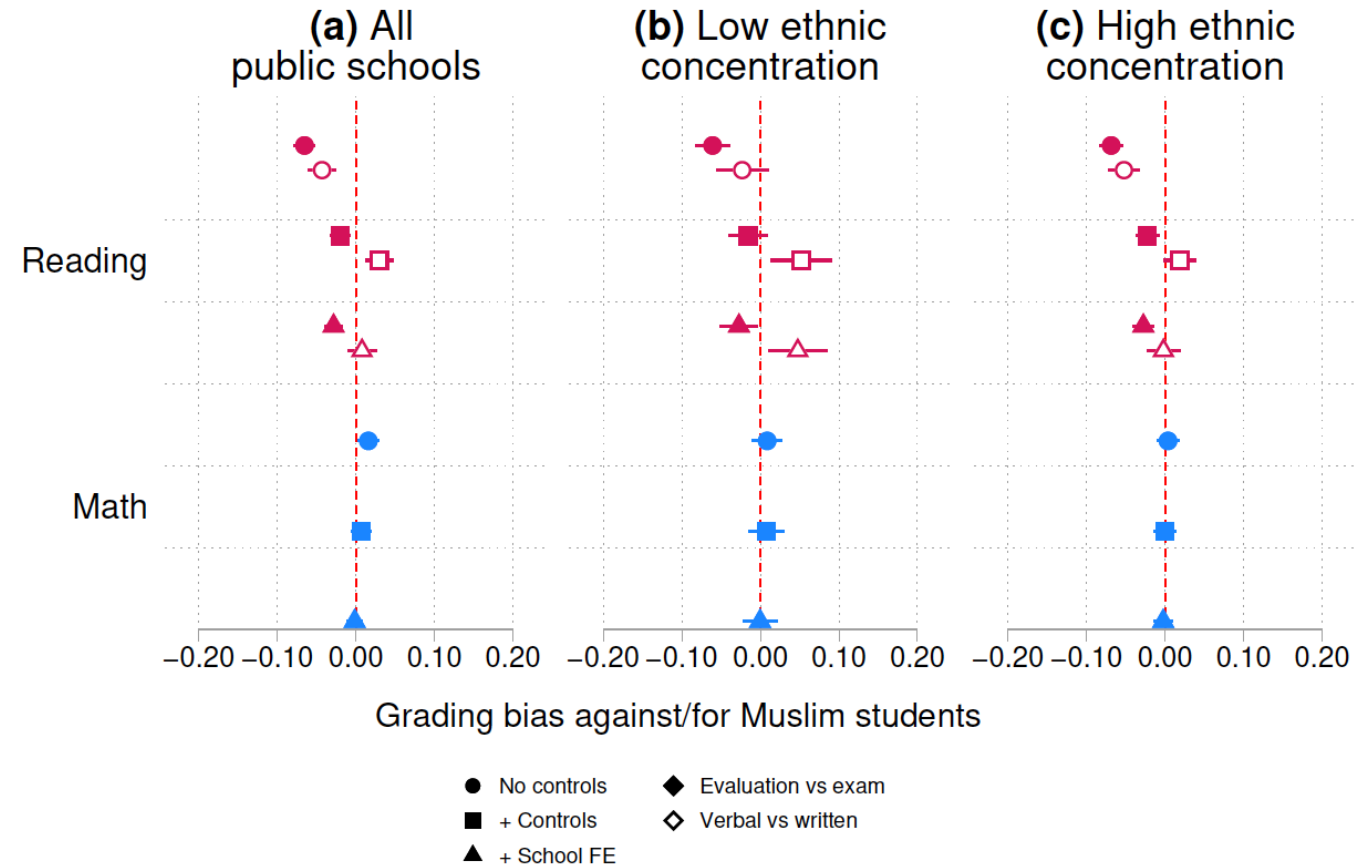
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- Sammenligneligt med effekter i Holland, og katolske skoler i USA (0.20-0.35 SD)
- Effekter tiltager over tid → *Ikke* skoleskift
- (+ mange andre robusthedsanalyser)



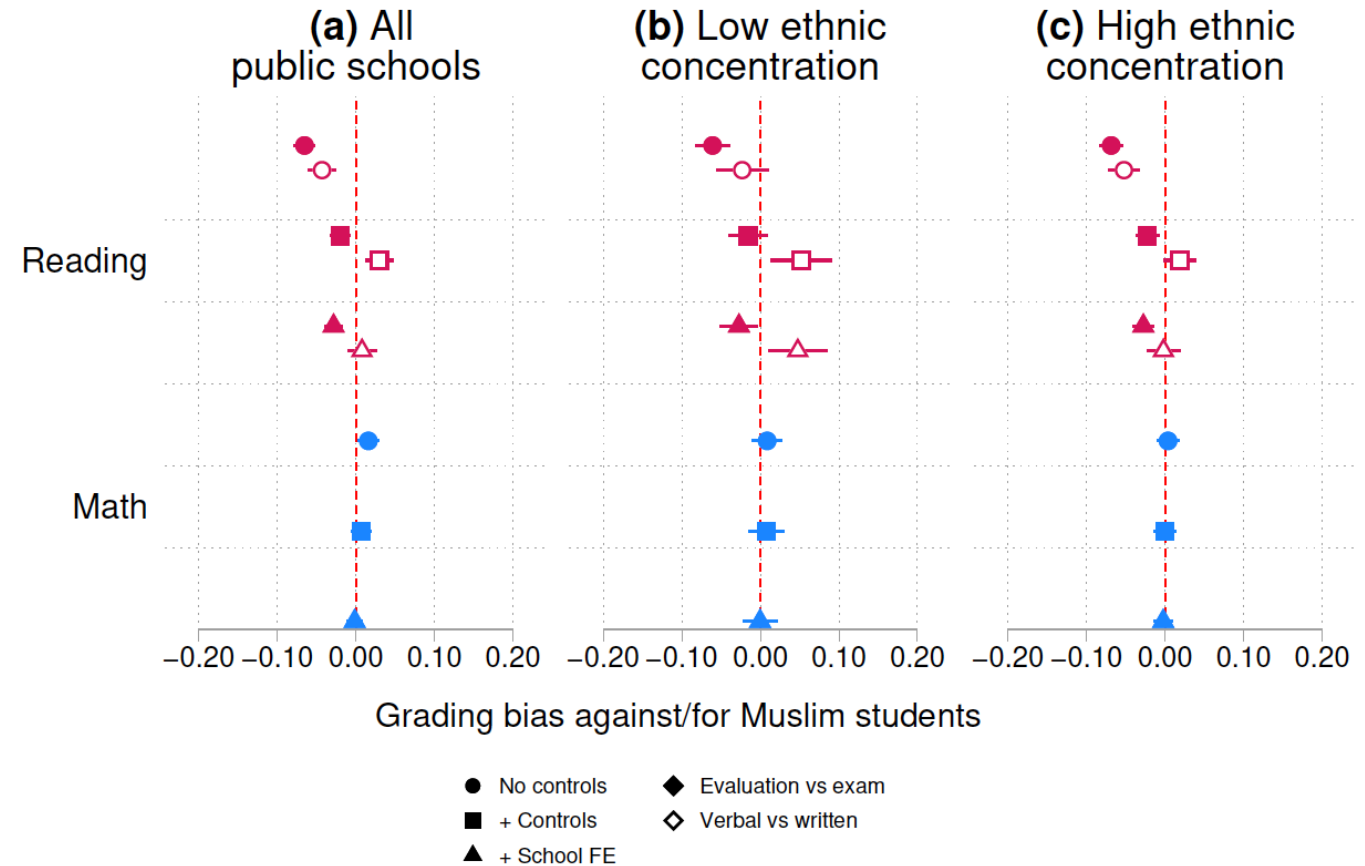
Lærerbias?

- Forskel mellem standpunkt og afgangseksamen



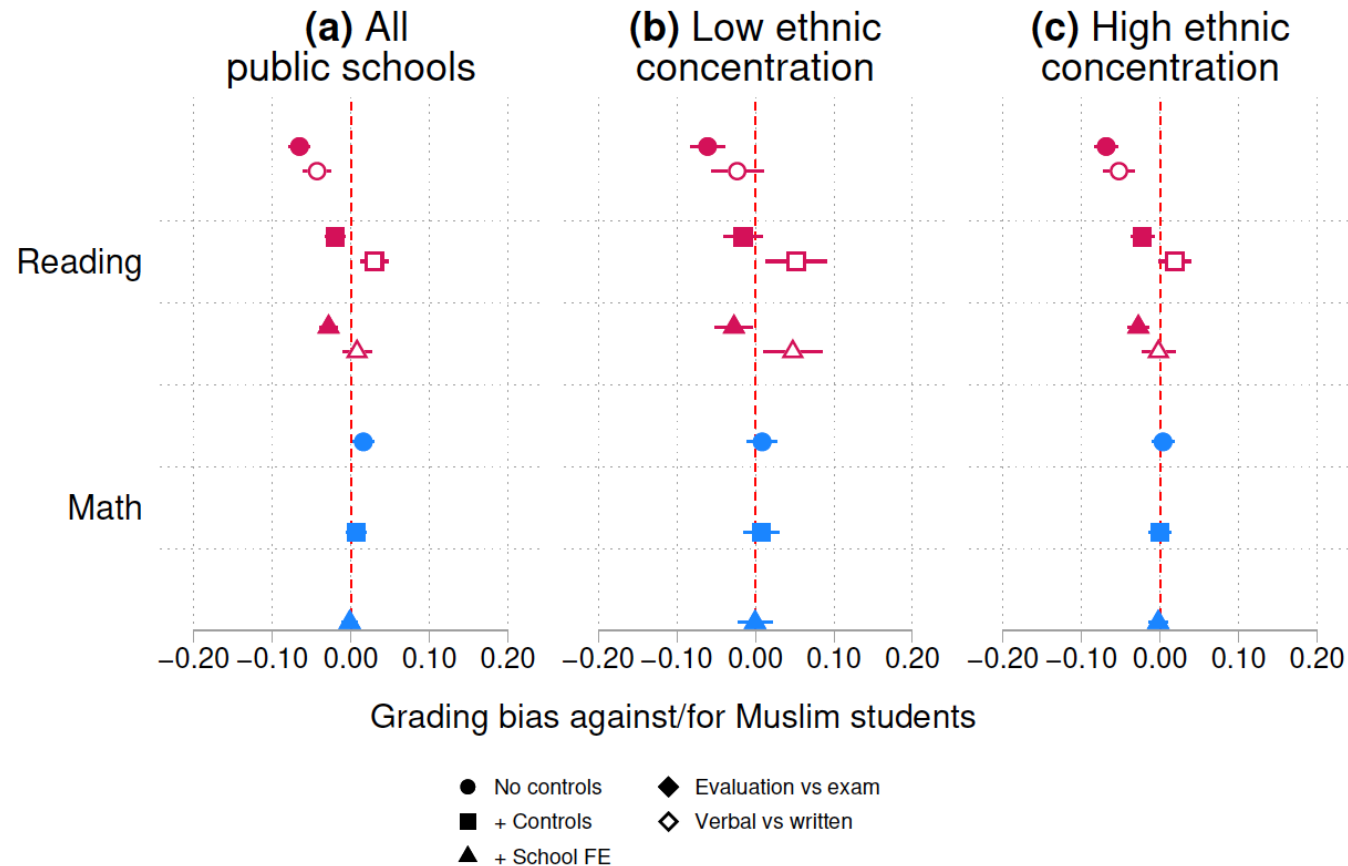
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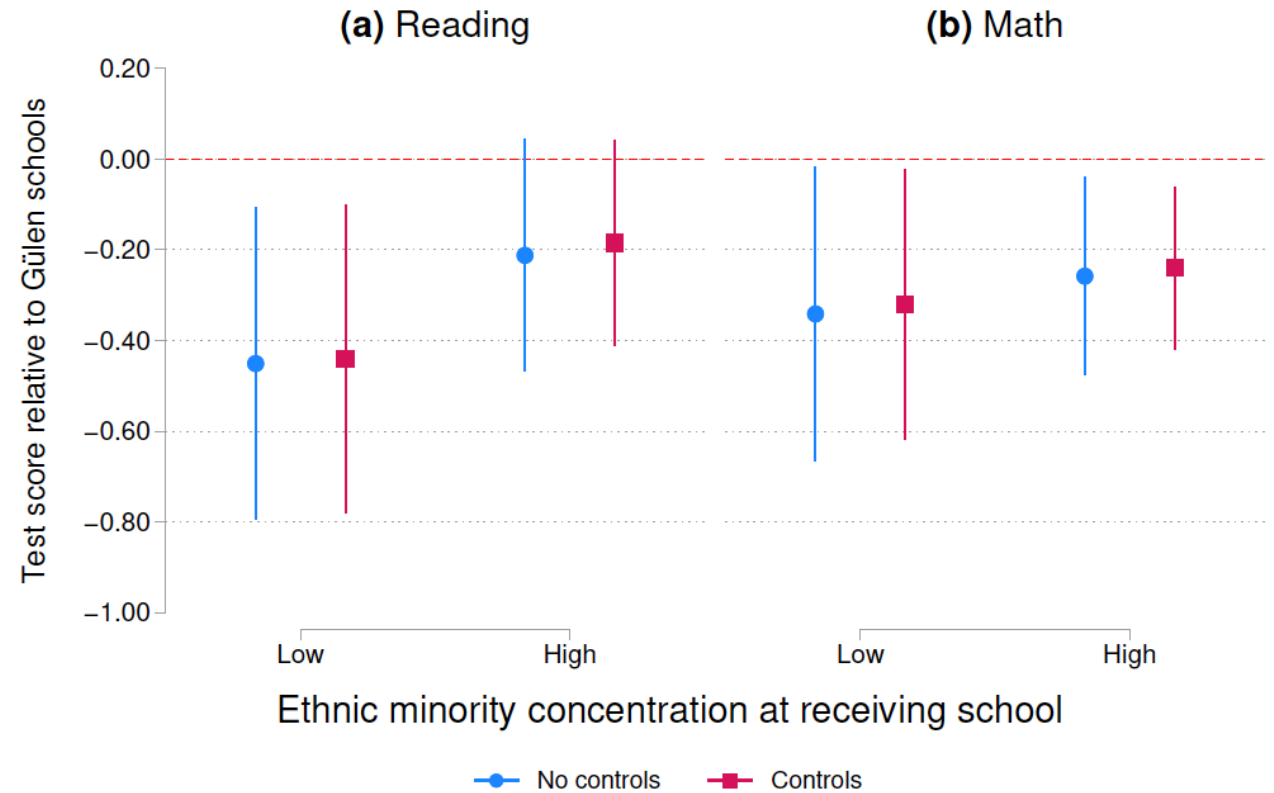
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- (Andre former for bias/diskrimination kan forekomme)



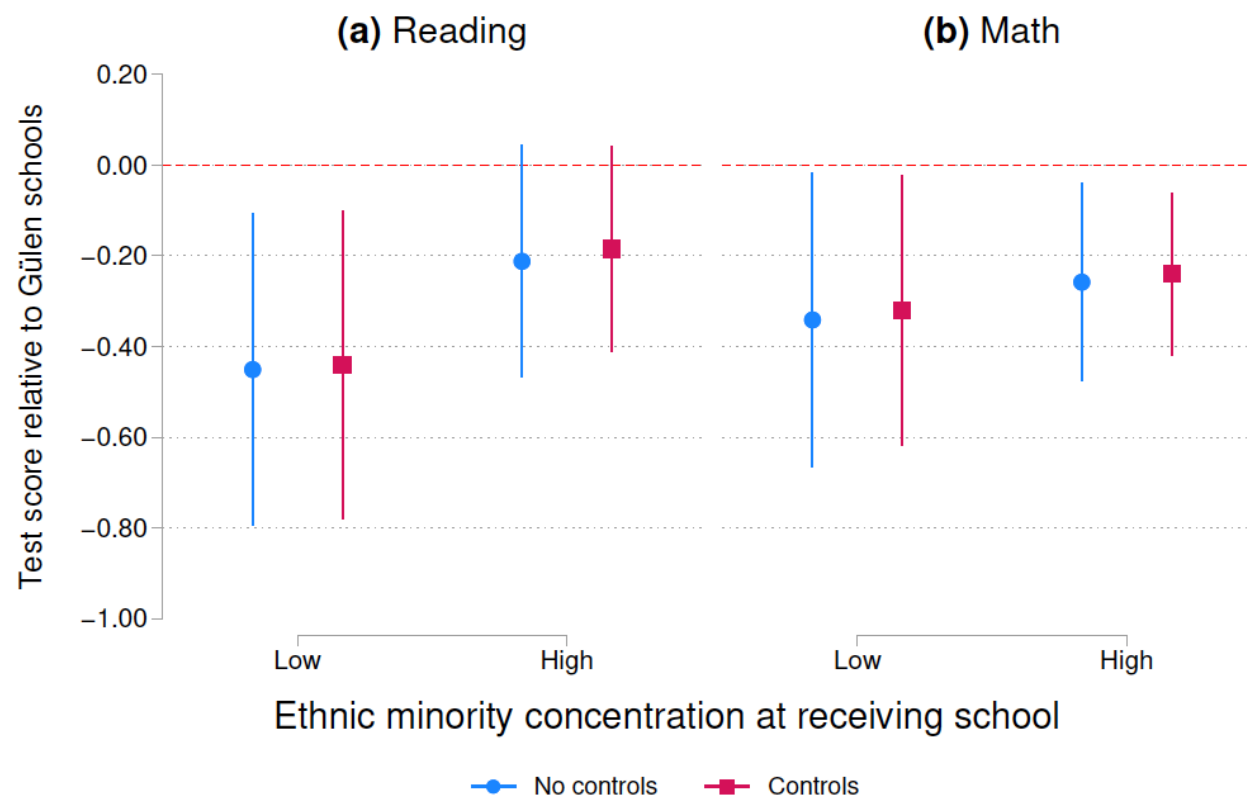
Etniske sammensætningseffekter

- Minoritetsbørn der flytter til skoler med en **højere** andel minoritetsbørn klarer sig **bedre**



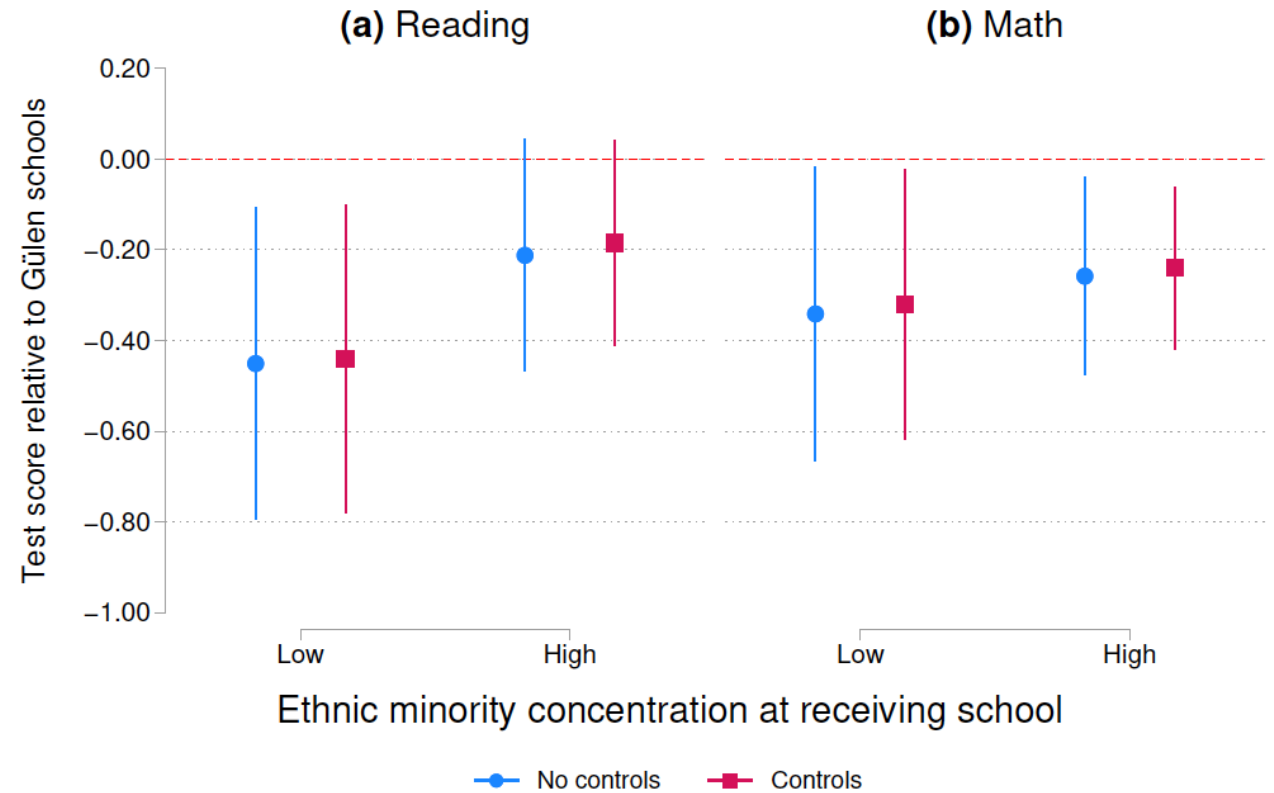
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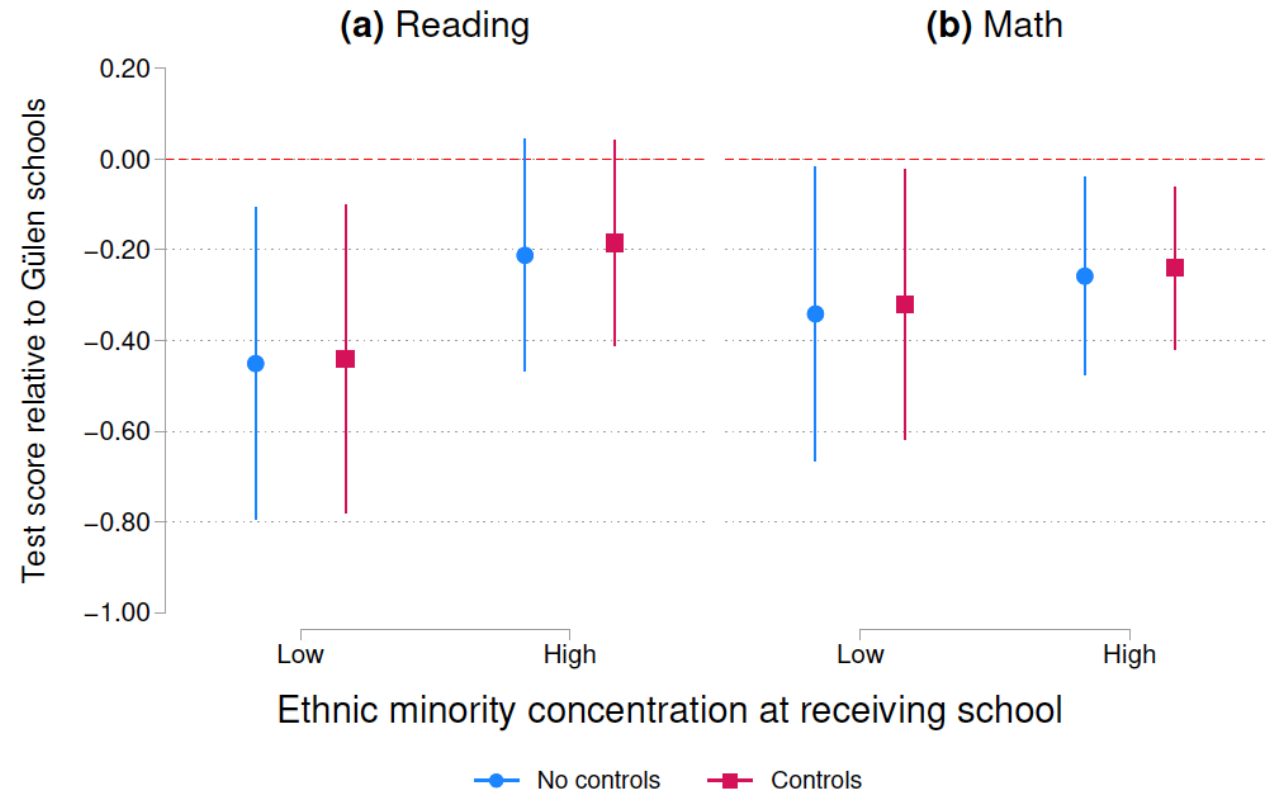
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- Forskning fra USA: "Teacher like me" og "Minority-serving institutions"



Trade off mellem læring og integration?

	High school outcomes			Social benefits recipient	
	Admission (age 18)	Completion (age 20)	Standardized GPA	Ever	Weeks
	(1)	(2)	(3)	(4)	(5)
Muslim school	0.113** (0.037)	0.122* (0.058)	0.355* (0.149)	-0.048 (0.033)	-6.397 (4.828)

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- Rockwool survey af Mouritsen et al. 2023

Tak for jeres opmærksomhed

www.saidhassan.net

APPENDIX – COVARIATE BALANCE

Table 1 Balancing test of covariates: Summary statistics and t -tests of differences in means, by treatment status.

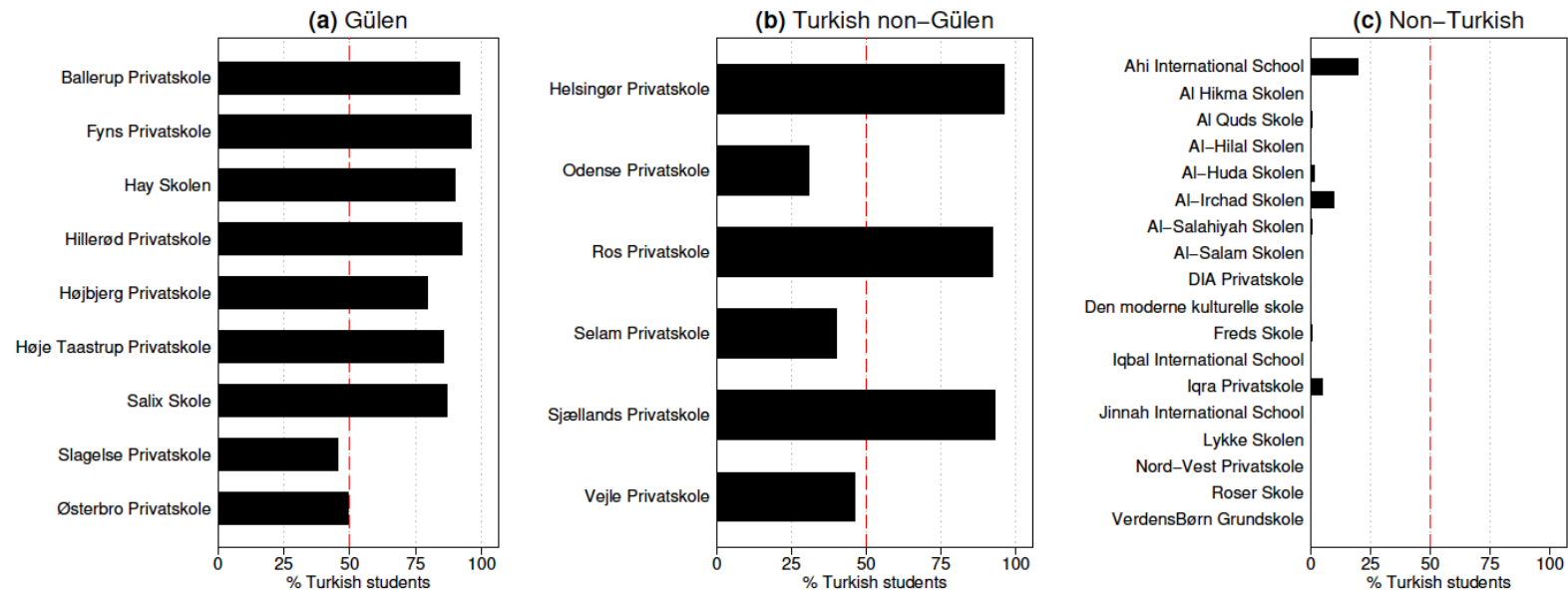
	Treatment		Control		t -test	
	Mean	(SD)	Mean	(SD)	Difference	p -value
Female	0.563	(0.496)	0.488	(0.501)	0.075	0.086
Age	16.153	(0.653)	16.226	(0.501)	−0.073	0.182
Both parents born in DK	0.815	(0.388)	0.774	(0.420)	0.041	0.236
Birth weight (kg)	3.412	(0.548)	3.413	(0.645)	−0.001	0.978
Parents divorced	0.080	(0.271)	0.071	(0.258)	0.008	0.726
Parental education (years)	12.823	(2.605)	12.644	(2.549)	0.179	0.436
Parental income	71.059	(30.889)	73.088	(25.237)	−2.028	0.438
Parent unemployed	0.198	(0.399)	0.263	(0.442)	−0.066	0.070
<i>Observations</i>	552		168			

APPENDIX – BACKGROUND INFO

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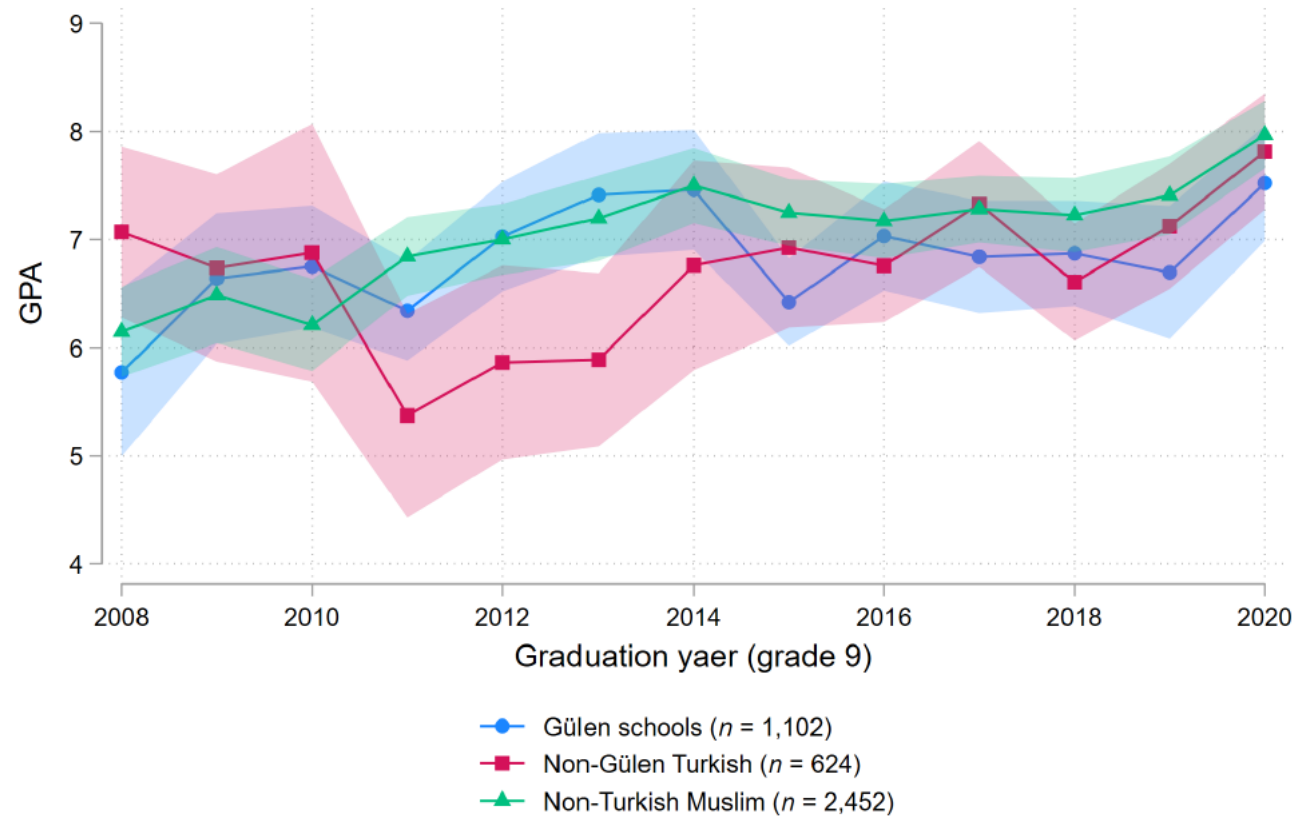
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Figure B3 Academic performance by Muslim school type and academic year. Raw average GPA in compulsory grade 9 subjects: reading, math, English, and science.



APPENDIX – DISRUPTION 1

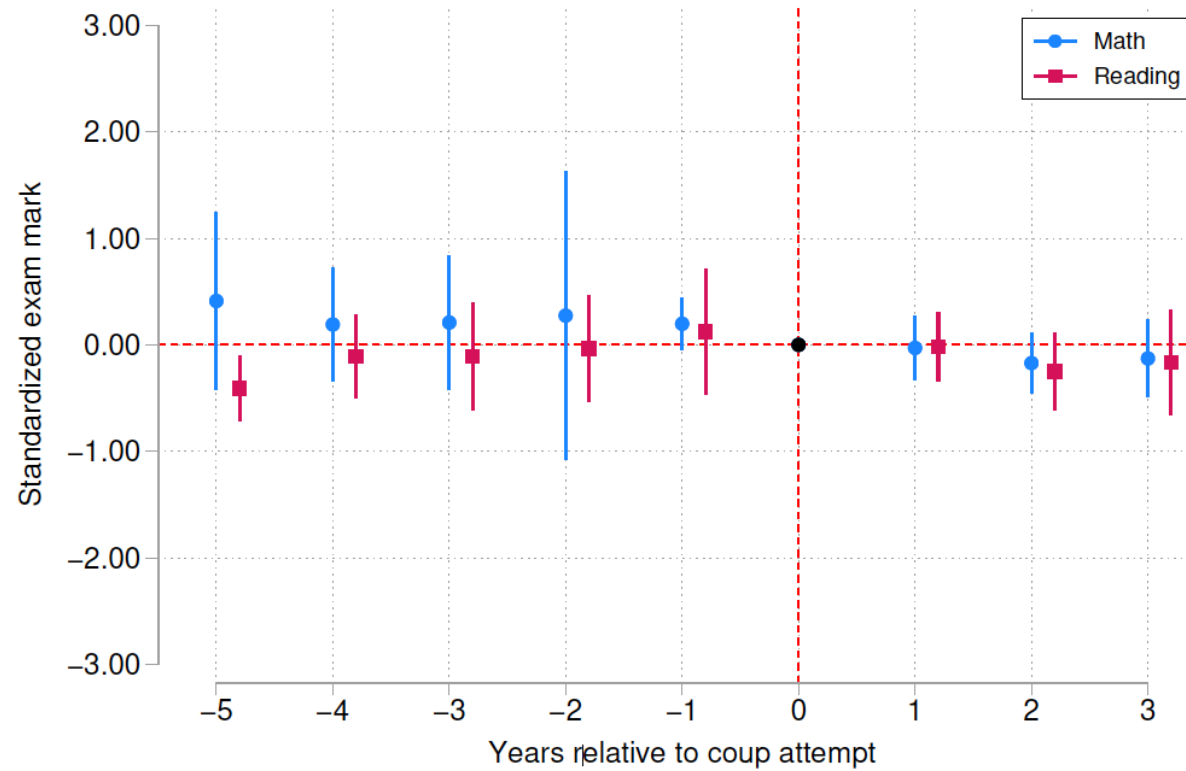


Figure 4 Event study estimates of test score dynamics in Gülen-associated Muslim schools relative to the coup attempt year.

APPENDIX – DISRUPTION 2

Table 3 Benchmark estimates of disruption effects associated with school transitions on students' test score performance.

	Full sample			Muslim children		
	(1)	(2)	(3)	(4)	(5)	(6)
$\delta: C_i$ (Change)	-0.104*** (0.010)	-0.096*** (0.009)	-0.080*** (0.008)	-0.079*** (0.022)	-0.067** (0.021)	-0.063*** (0.019)
$\alpha: y_{i,t-1}$	0.683*** (0.003)	0.615*** (0.003)	0.601*** (0.003)	0.544*** (0.006)	0.504*** (0.006)	0.493*** (0.006)
$\gamma: \bar{y}_{-i,t}$			0.381*** (0.008)			0.338*** (0.018)
<i>Observations</i>	373,575	373,574	373,574	35,076	35,076	35,076
R^2	0.431	0.496	0.510	0.365	0.421	0.437
Covariates	No	Yes	Yes	No	Yes	Yes

Note: Full population data on all students enrolled in Danish public schools from 2010–2018 in grades 8–9. The dependent variable, y_{it} , is reading test scores in grade 9. The variables $y_{i,t-1}$ and $\bar{y}_{-i,t}$ capture lagged test scores and the destination school's average GPA excluding the focal student, respectively. All test scores are standardized to mean 0 and standard deviation 1 in the full sample. For further details on data and measures, see [Appendix D](#). Standard errors in parentheses are clustered at the destination school. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

APPENDIX – DISRUPTION 3

Table 4 Association between school transitions and test score performance, by school type transitions. *Dep. var.*: standardized written reading exam in compulsory school's 9th grade.

	From public school			From private school		From Muslim school	
	To public (1)	To private (2)	To Muslim (3)	To private (4)	To public (5)	To Muslim (6)	To public (7)
Moved	−0.139*** (0.009)	−0.049*** (0.014)	0.217*** (0.038)	−0.130*** (0.021)	−0.155*** (0.013)	0.132 (0.088)	−0.217** (0.063)
Constant	−0.185*** (0.004)	−0.198*** (0.004)	−0.199*** (0.004)	−0.088*** (0.009)	−0.070*** (0.010)	−0.026 (0.079)	−0.039 (0.083)
<i>N</i> (total)	256,684	265,405	265,405	43,135	46,143	1,565	1,648
<i>N</i> (movers)	33,833	9,071	113	2,133	4,474	45	269
<i>R</i> ² adj.	0.218	0.214	0.214	0.193	0.193	0.159	0.155

Note: “Private” refers to all non-Muslim independent schools in Denmark. “Muslim” refers to Muslim independent schools. Transitions are measured between grades 7 and 9, i.e. in the compulsory school's lower secondary stage in Denmark (“udskoling”). For simplicity, I drop students who move more than once during this period. The main independent variable “Move” is a dummy indicating that the student moved school either within the same school type (e.g., public to public in column 1) or to a different school type (e.g., public to non-Muslim private in column 2). Two types of transitions are excluded due to insufficient sample sizes for confidentiality reasons (those moving from non-Muslim private to Muslim and from Muslim to non-Muslim private). Data from 2010–2015 to avoid conflating moves with the coup attempt. Standard errors in parentheses are clustered at the destination school. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

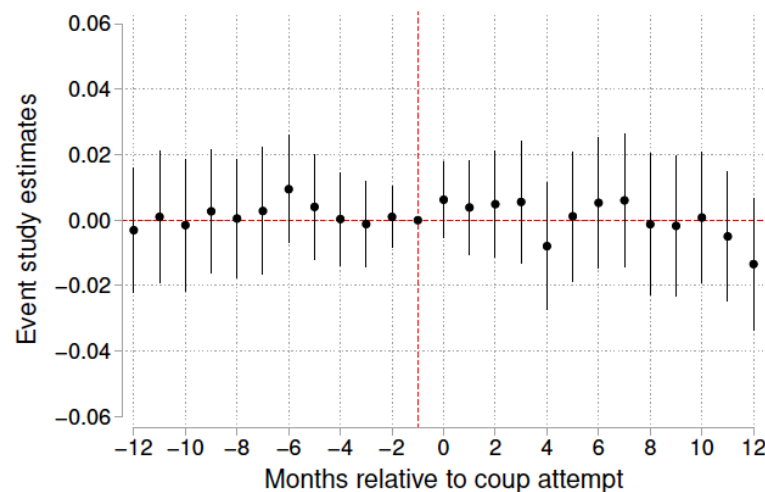
APPENDIX – DISRUPTION 4

Table 5 Disruption analysis: Using movers from Gülen to non-Gülen Muslim schools to identify the learning loss associated with school changes.

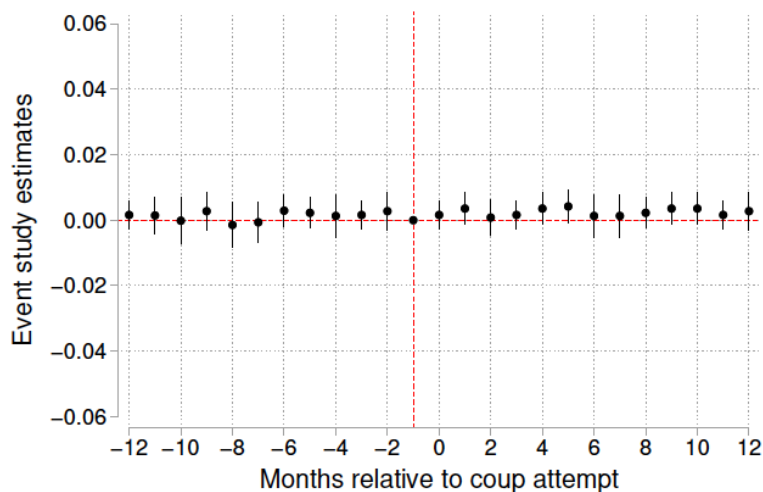
	Reading		Math	
	(1)	(2)	(3)	(4)
Coefficient	−0.184	−0.018	−0.088	0.065
95% CIs	[−0.525,0.156]	[−0.348,0.312]	[−0.469,0.292]	[−0.357,0.488]
<i>p</i> -value	(0.264)	(0.910)	(0.624)	(0.743)
<i>Observations</i>	663	663	649	649
<i>R</i> ²	0.022	0.144	0.015	0.089
Covariates	No	Yes	No	Yes

Note: This table reports regressions where the independent variable is a dummy identifying those who moved from a Gülen-associated Muslim school to a non-Gülen Muslim schools (n (movers) = 38), following the coup attempt, to identify disruption effects related to changing schools, net of the school environment. 95% CIs in brackets, and p -values in parentheses. Standard errors clustered at the school level.

APPENDIX – DISRUPTION 5



(a) Sick leave



(b) Divorce

Figure 6 Event study estimates of the impact of the coup attempt on parents with children enrolled in Gülen vs non-Gülen Turkish Muslim schools.

$$S_{it} = \sum_{k \neq -1} \alpha_k \mathbb{1}\{r_{it} = k\} + \sum_{k \neq -1} \delta_k (\mathbb{1}\{r_{it} = k\} \times D_i) + \gamma D_i + \mathbf{x}'_i \boldsymbol{\beta} + \varepsilon_{it}, \quad (6)$$

APPENDIX – ETHNIC SIMILARITY 1

Table 6 Differences in peer and teacher characteristics between low- and high ethnic concentration receiving schools.

	Low Ethnic Concentration		High Ethnic Concentration		<i>t</i> -test	
	Mean	(SD)	Mean	(SD)	Difference	<i>p</i> -value
<i>School and peer characteristics</i>						
Ethnic minority	0.274	(0.094)	0.603	(0.174)	−0.329	0.000
School GPA (std)	−0.130	(0.267)	−0.347	(0.248)	0.217	0.000
School cohort size	188.013	(50.768)	167.988	(56.120)	20.025	0.019
Paternal income	59.919	(9.740)	47.384	(6.937)	12.535	0.000
Maternal income	47.152	(4.703)	38.164	(4.241)	8.988	0.000
Paternal education	13.969	(0.920)	13.020	(0.500)	0.949	0.000
Maternal education	13.793	(1.003)	12.495	(0.910)	1.298	0.000
<i>Teacher characteristics</i>						
Ethnic minority	0.053	(0.050)	0.115	(0.076)	−0.062	0.000
Age	42.541	(2.716)	42.289	(1.702)	0.252	0.481
Certified	0.801	(0.093)	0.795	(0.087)	0.006	0.677
Experience (years)	12.678	(2.350)	12.566	(1.735)	0.111	0.733
Experience ≥ 3 years	0.887	(0.064)	0.879	(0.057)	0.008	0.390
<i>Observations</i>	77		83			

APPENDIX – ETHNIC SIMILARITY 2

Table 7 Association between ethnic minority status and educational aspirations and the moderating role of ethnic concentration at the school level.

	<i>Dep. var.: Educational aspirations</i>			
	(1)	(2)	(3)	(4)
Minority student	0.006 (0.012)	0.097*** (0.018)	0.108*** (0.022)	0.075*** (0.020)
% Minority in school			0.000 (0.000)	
Minority student $\times$ % Minority in school			−0.001 (0.001)	0.000 (0.000)
Constant	0.794*** (0.005)	0.748*** (0.005)	0.741*** (0.007)	0.749*** (0.003)
<i>Observations</i>	46,988	44,105	36,212	36,209
<i>R</i> ² adj.	0.000	0.197	0.198	0.251
Covariates	No	Yes	Yes	Yes
School fixed effects	No	No	No	Yes

Note: “Minority student” is a dummy variable indicating whether the student is either an immigrant or descendant from a non-Western country. “% Minority in school” is the percentage of non-Western immigrants or descendants in the school. In column (4) this measure is absorbed into the school fixed effect. The number of observations decreases from (1) to (2) due to missings in the covariates; from (2) to (3) due to missing school-level counts of minority students (for data confidentiality); and from (3) to (4) due to singleton observations after the introduction of school fixed effects. Standard errors are clustered at the school level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

APPENDIX – GRADING BIAS VS SUBTLE BIAS

Table 8 Teacher biases against/for students from Muslim-majority countries in evaluations of their academic, social, and personal readiness for high school. Danish public schools in 2015–2016.

	<i>Dep. var.: Teacher-evaluated student readiness for high school</i>					
	Academic readiness		Social readiness		Personal readiness	
	(1)	(2)	(3)	(4)	(5)	(6)
Muslim student	−0.154*** (0.008)	−0.030*** (0.008)	−0.133*** (0.009)	−0.041*** (0.009)	−0.181*** (0.009)	−0.057*** (0.009)
Constant	0.829*** (0.003)	0.784*** (0.003)	0.892*** (0.003)	0.832*** (0.004)	0.831*** (0.003)	0.746*** (0.005)
<i>Observations</i>	94,992	94,992	73,830	73,830	73,830	73,830
<i>R</i> ² adj.	0.012	0.240	0.016	0.106	0.020	0.149
Covariates	No	Yes	No	Yes	No	Yes

Note: All regressions are linear probability models where the outcome is a binary evaluation of whether the teacher evaluates the student to be ready for high school along three dimensions (academic, social, and personal). For further details, see [Appendix F](#). Standard errors are clustered at the school level. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.